

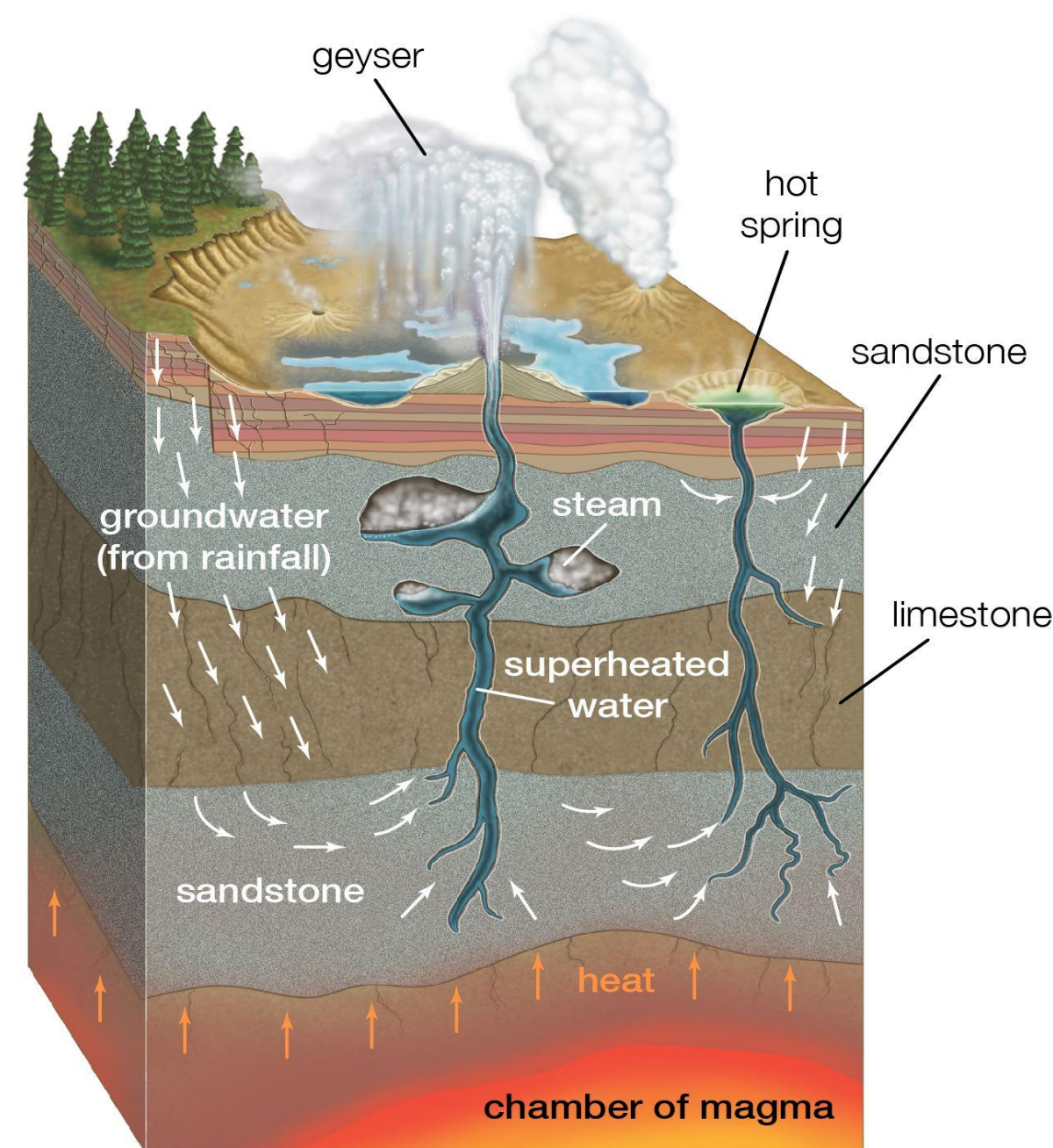
# Designing a Geyser Eruption Monitoring Device for Physical Modeling Setup

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## Introduction

- A geyser (*Fig. 1*) can be defined as a periodic eruptive spring resulting from Earth's thermal activity (Brandendourger et al, 2016).



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Figure 1: Example of a Geyser and its plumbing system.

## Methods

### Setup

- Allowed the heating plate to heat up for five minutes at level 5 (201 C°).
- Added 100cc of water to the reservoir.
- Placed the geyser physical model on top of the heating plate to begin the experiment.
- The monitoring device collected 1 data point every 5 seconds.
- The thermocouple was placed in between the reservoir and the heating plate. It displayed the temperature of the location constantly, but it had to be recorded manually.
- The heating power was kept constant for the first hour. Then, to test the capabilities of the setup the heating level was changed to 3 (152 C°) for 20 minutes, and level 6 (355 C°) for the last 20 minutes of the experiment.

## Results

- Successfully monitored and recorded multiple parameters of the geyser physical modeling setup (*Fig. 3a-3c*).

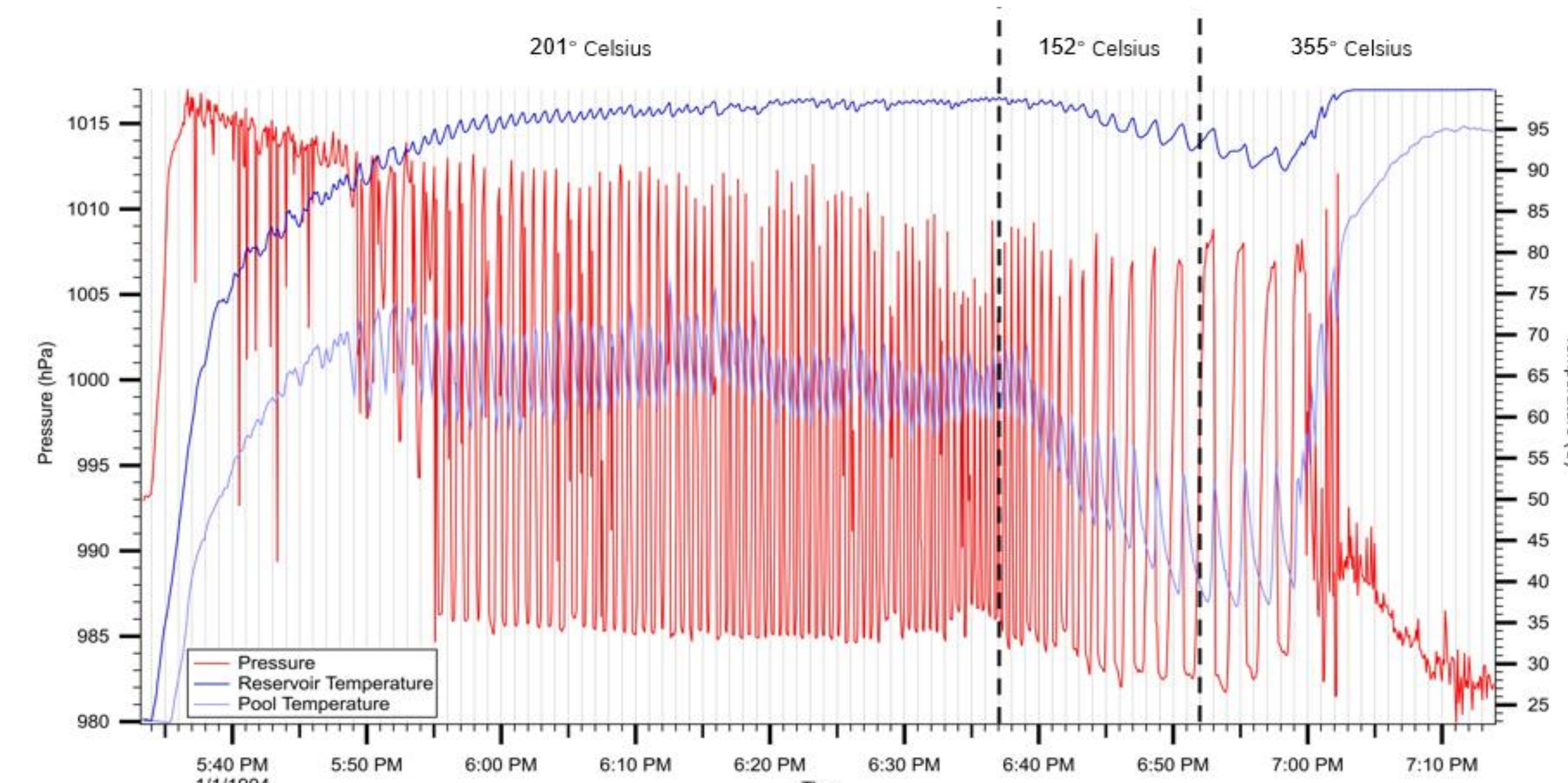


Figure 3a: Data from all the whole experiment.

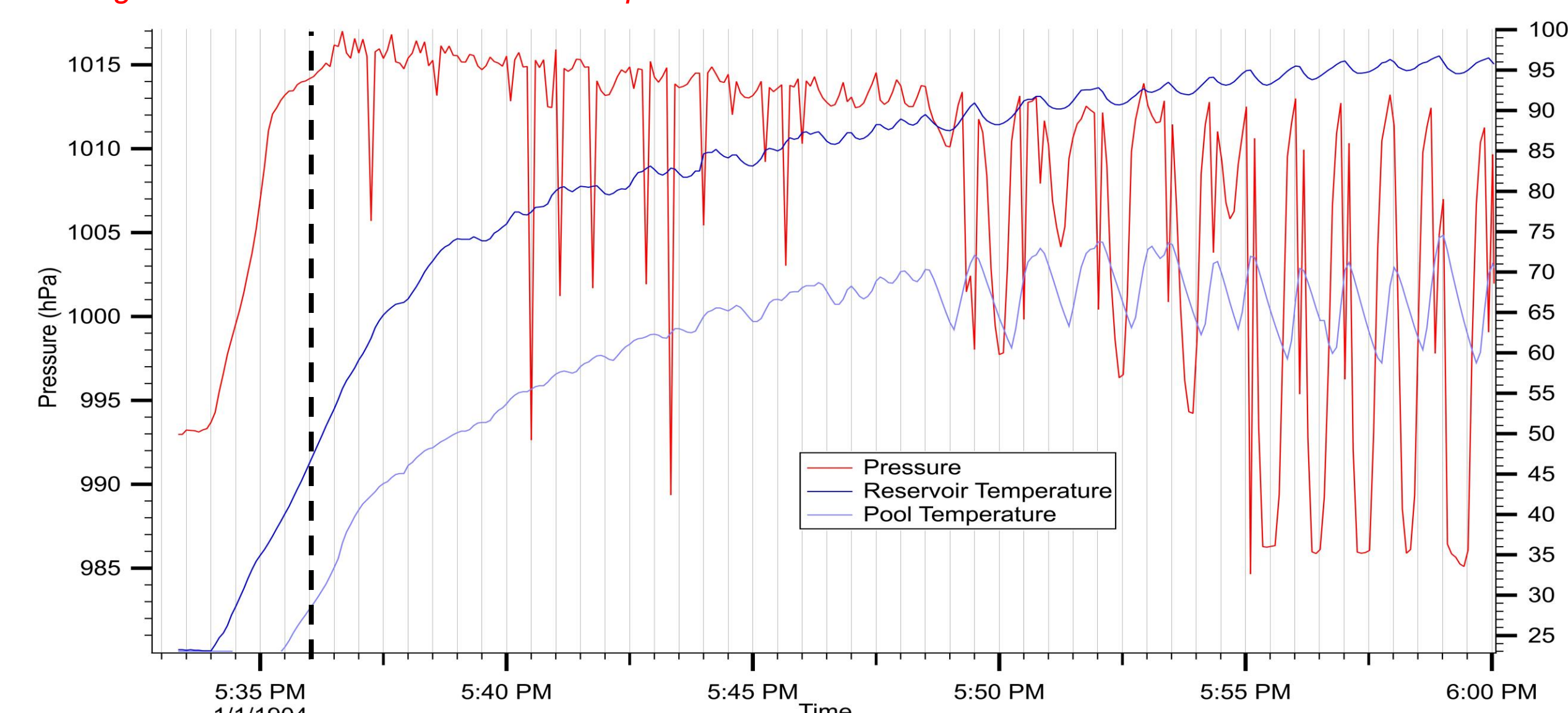


Figure 3b: Data from the beginning of the experiment. The black dashed line indicates the time of the first eruption.

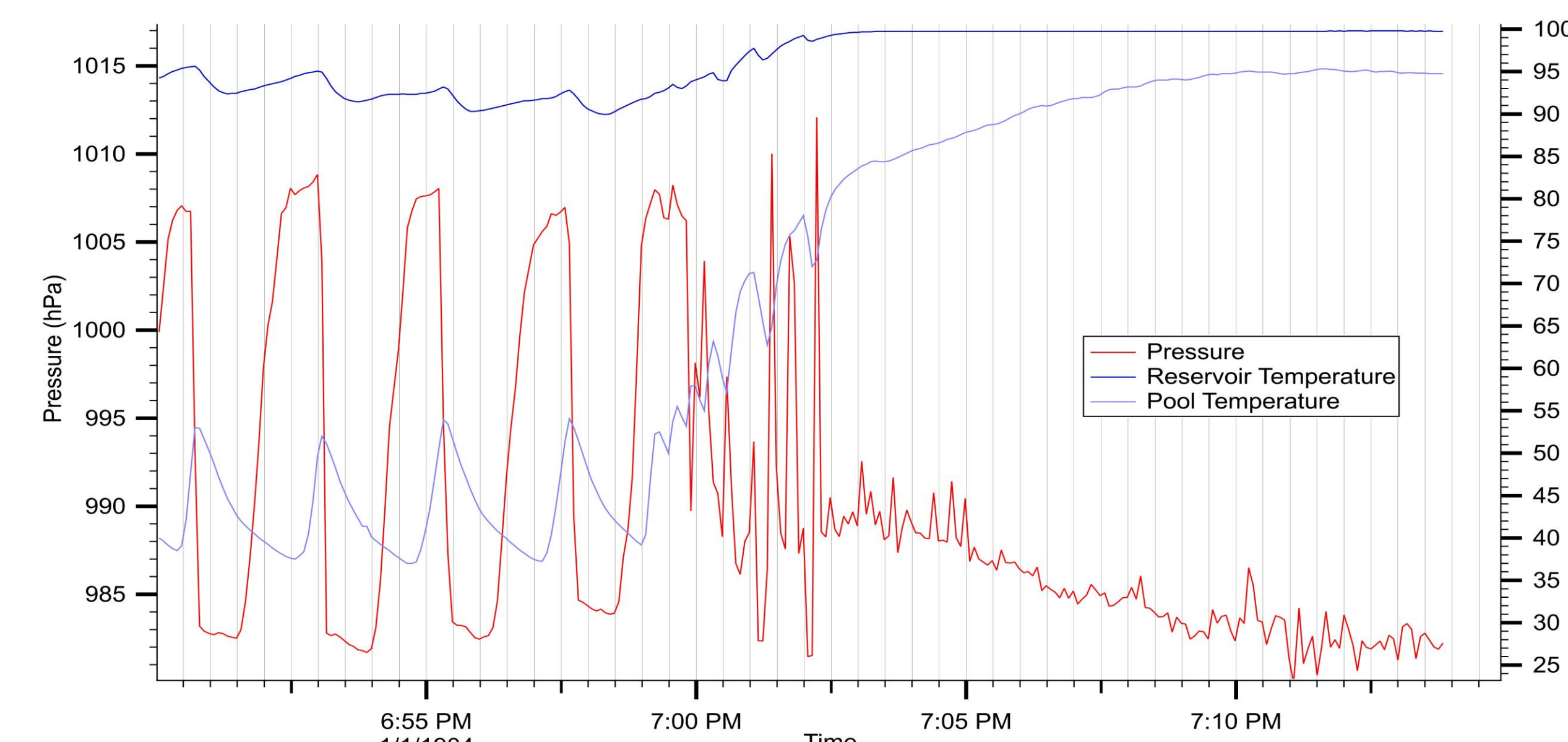


Figure 3c: Data from the end of the experiment. The black dashed line indicates when the heat setting on the heating plate was changed from 3 (152 C°) to 6 (355 C°).

## Fieldwork

- The time between two eruptions was recorded (*Fig. 4a*).

### Observations

- Increased activity in the pool began as soon as 6 minutes before the main eruption.
- The mini eruption caused a longer eruption wait time (*Fig. 4c*). This could be attributed to a loss in pressure in the system.

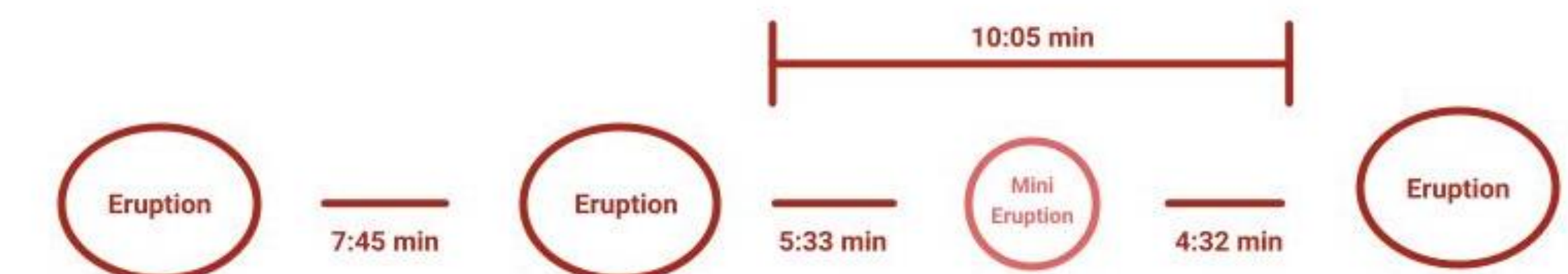


Figure 4a: Timeline of recorded eruptions.



Figure 4b: Strokkur eruption.



Figure 4c: Strokkur mini eruption.

## After Iceland

- Investigate the anomalies observed in the data. Make changes to the monitoring device as necessary.
- Conduct more experiments. Test different temperatures, water volumes, and measurements of the physical model (reservoir, vent, and pool).
- Try to replicate the mini eruption observed.
- Desing a device that can closely monitor the activity in pool, as I believe this could provide better insights into the eruption timing.
- Modify the physical model to get closer to real-life scenarios (*Fig. 5*).

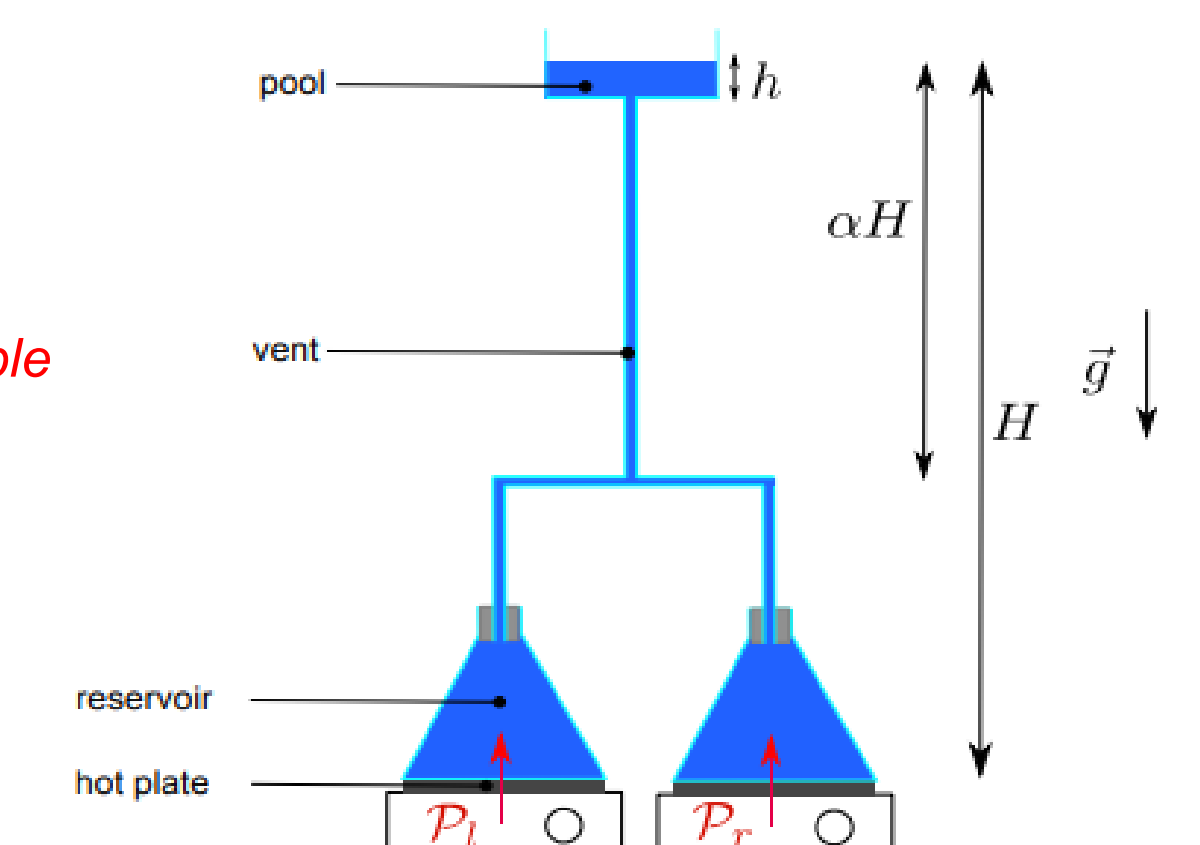


Figure 5: Example of a possible future physical model.

## Acknowledgments

Students who have previously conducted similar research: Kirsten Ruffert, Allison Balliet, Henry Rohde, Maria Zenteno.

UW Whitewater Undergraduate Research Program.

Brandendourger, M., Dorbolo, S., & Texier, B. D. (16 Mar, 2016). Physics of a toy geyser. *Cornell University*.

## Objectives

- Gain a better understanding of a geyser's mechanisms.
- Learn about the hardware and software necessary to build a geyser eruptions monitoring device.
- Design and build a monitoring device that can accurately record the pressure, temperature, and periodic eruptions of our physical model in real-time.

## Materials

### Monitoring Device (*Fig. 2a*)

- 1 Raspberry Pi
- 2 Temperature sensors
- 1 Pressure sensor
- 1 i2c amplifier
- 2 Moisture sensors

### Separate Device (*Fig. 2b*)

- Thermocouple

### Physical Model (*Fig. 2c*)

- Beaker
- Heating plate
- Tubing
- Funnel
- Stand with clamps

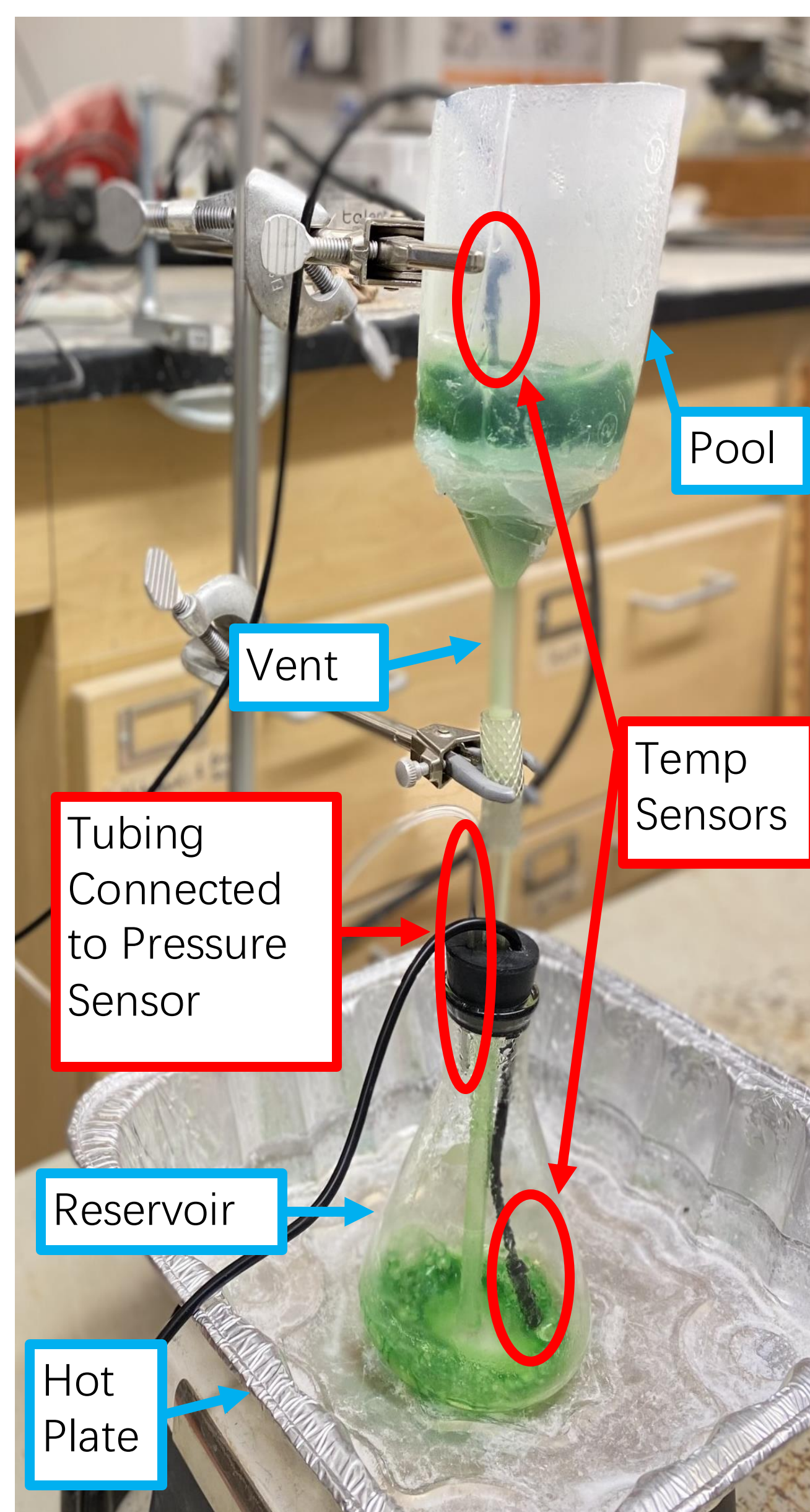


Figure 2c: Physical modeling setup.

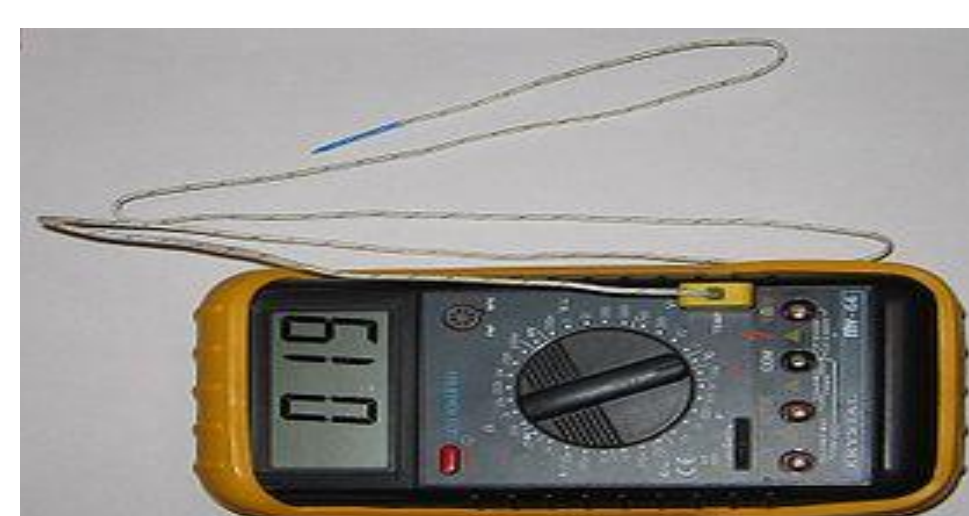


Figure 2b: Thermocouple.



Figure 2a: Monitoring device.